Royal Devon and Exeter NHS
NHS Foundation Trust

Clinical Guideline for
EXCHANGE TRANSFUSION

SUMMARY

This guideline outlines the treatment of hyperbilirubinemia by exchange transfusion

KEY POINTS

- The purpose of exchange transfusion
- The complications of exchange transfusion
- The equipment required for an exchange transfusion
- How to perform an exchange transfusion
- Post exchange care

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This clinical guideline is designed to provide advice, guidance and direction to staff whilst leaving room for professional judgement, adaptation and to fit individual circumstances.

1. INTRODUCTION

1.1 Exchange transfusions are used primarily for infants with Haemolytic Disease of the Newborn (HDN) to remove antibody coated red cells and bilirubin and to correct the anaemia associated with haemolysis.

1.2 The affected infant's blood is removed and replaced in small aliquots of compatible blood completing the transfusion by leaving the infant with a 5-10ml deficit.

1.3 Due to the significant decline in the incidence of HDN following the introduction of maternal anti-D Ig prophylaxis, the use of intense phototherapy and intravenous immunoglobulin treatment, exchange transfusion has become an uncommon procedure. Decision for exchange transfusion should be made by service/ on-call consultant with consideration of early discussion with a Level 3 unit.

2. BACKGROUND

2.1 Exchange transfusion works by lowering serum bilirubin levels and reduces the risk of kernicterus and potential associated brain damage.

2.2 It removes the infant's sensitised red blood cells and the circulating antibodies and to reduce the degree of red cell destruction.

2.3 It can correct serious anaemia and increase the oxygen capacity of the infant's blood. Alleviate sepsis or disseminating intravascular coagulation.

3. INDICATIONS

3.1 Use a double volume exchange transfusion to treat babies; whose serum bilirubin level indicates its necessity (see threshold table below) and/or with clinical features of acute bilirubin encephalopathy.

3.2 All babies who meet exchange transfusion threshold will need to be discussed with service/ on-call consultant urgently.

Consensus-based bilirubin thresholds for the management of babies of 38 weeks or more gestational age with hyperbilirubinaemia

Age (hours)	Bilirubin measurement (micromol/litre)			
0			> 100	> 100
6	> 100	> 112	> 125	> 150
12	> 100	> 125	> 150	> 200
18	> 100	> 137	> 175	> 250
24	> 100	> 150	> 200	> 300
30	> 112	> 162	> 212	> 350
36	> 125	> 175	> 225	> 400
42	> 137	> 187	> 237	> 450
48	> 150	> 200	> 250	> 450
54	> 162	> 212	> 262	> 450
60	> 175	> 225	> 275	> 450
66	> 187	> 237	> 287	> 450
72	> 200	> 250	> 300	> 450
78		> 262	> 312	> 450
84		> 275	> 325	> 450
90		> 287	> 337	> 450
96+		> 300	> 350	> 450
Action	↓ Repeat bilirubin measurement in 6–12 hours	↓ Consider phototherapy and repeat bilirubin measurement in 6 hours	↓ Start phototherapy	↓ Perform an exchange transfusion unless the bilirubin level falls below threshold while the treatment is being prepared

4. PREPARATIONS FOR AN EXCHANGE TRANSFUSION

4.1 Ensure parents are fully informed and have consented to the procedure. It is the medical staff's responsibility to obtain consent**.**

4.2 Infant should be admitted to ITU**.**

4.3 Maintain a neutral thermal environment by nursing in an incubator or cosy cot.

4.4 Commence continuous monitoring of heart rate, respiratory rate, blood pressure, temperature and oxygen saturations. Baby should also be on a cardiac monitor. Record baseline before procedure and then every 15 mins once procedure **starts.**

4.5 Gain access via umbilical vein and umbilical artery as per unit guidelines.

4.6 Acceptable access alternatives.

IN	OUT
Umbilical venous catheter (UVC)	**Umbilical arterial catheter (UAC)**
UVC	**Peripheral intravenous catheter (PIV)**
PIV	**UVC**
PIV	**UAC**
PIV	**Peripheral arterial line (PAL)**
UVC	**UVC**

4.7 **Analgesia** as per unit policy.

4.8 Ensure baby is NBM 2 hours prior to procedure**.**

4.9 Insert nasogastric tube.

4.10 Commence intravenous fluids.

4.11 Take blood for new born blood spot day 0, pre-exchange transfusion blood samples, including blood glucose.

4.12 Prepare equipment using aseptic technique.

5. EQUIPMENT REQUIRED

- IV volumetric pump
- Blood filter giving set
- Ranger blood warming machine
- Ranger blood warming set
- Trolley
- Sterile field
- Green luer lock connector
- Closed system urine bag
- Luer lock syringes 50 ml, 10 ml, 5 ml, 2 ml.
- 3 way tap x 2
- Nacl ampules
- Exchange transfusion blood in/out chart (Appendix 1, please use SmartPhrase .nnuexchangetransfusion to record on MyCare)
- Full blood count bottles x 3
- Gold tops x 3
- Clotting bottles x 3
- Blood culture bottle x 1
- 10% Calcium Gluconate

- Clock

6. BLOOD PRODUCTS

6.1 Adhere to Trusts Blood Transfusion Policy

6.2 Volume of blood required for exchange is **160 mls/kg** (twice circulating blood volume)

6.3 Blood should be used in order of expiry

6.4 Blood should be less than 5 days old, ABO compatible with maternal/neonatal sample

6.5 Irradiated within 24 hours

6.6 Used within 24 hours

6.7 CMV negative

6.8 Plasma reduced

6.9 Report any side effects/abnormal observations immediately

7. PROCEDURE

7.1 Check blood according to Trusts Blood Policy.

7.2 Insert the Ranger warming set into the Ranger warming machine prior to running through the blood. Connect the inlet line to the blood set. Invert the bubble trap and prime the inlet line and bubble trap until full. Turn the bubble trap right side up and prime the patient line.

Close the patient line clamp and turn the warming unit on. It is now ready for use. The warmer will automatically heat up to 41 degrees. If the unit alarm sounds turn the unit off and seek medical electronics advice.

7.3 Connect the patient line to the 3 way tap on the venous line**.**

7.4 Connect the green luer lock connector to the arterial line and closed-circuit urine bag.

7.5 Minimum staffing for this procedure is a registrar/ senior SHO and neonatal nurse. All clinical staff should be ready to start the procedure **simultaneously and should have no other workload.** Exchange transfusions should only be done with presence of service/ on-call consultant on the unit.

7.6 Medical staff should be responsible for removing and infusing the blood. A double-volume exchange should take 1.5 – max 3 hours. Each cycle of in/ out should take roughly 4 minutes (‘out’ 1 minute, ‘in’ 1 minute and ‘pause’ 1-2 minutes).

7.7 The size of aliquot can be calculated based on weight, using the table below.

Weight (kg)	Aliquot size (ml)
<1kg	5ml
1-2kg	10ml
2-3kg	15ml
>3kg	20ml

7.8 Blood should always be **removed first**. Blood can then be removed and replaced in synchronisation, until the prescribed amount of blood has been exchanged.

7.9 At the end of the procedure the infant's fluid balance should be in deficit by 1 aliquot.

7.10 Blood intake and output should be recorded on the exchange transfusion recording chart.

7.11 Ionised calcium should be checked halfway through exchange transfusion. At the discretion of the medical staff 1ml calcium gluconate 10% IV may be given slowly for every 100mls of exchanged blood. This should be given through a separate IV line to the blood.

8. BLOOD SAMPLES REQUIRED

PRE EXCHANGE	**FBC** **U&E'S** **SBR** **BLOOD GAS** **HAEMATOCRIT** **TRUE GLUCOSE** **COAGULATION PROFILE** **BLOOD CULTURE** **DAY O BLOOD SPOT**
HALF WAY THROUGH EXCHANGE	**FBC** **U&E'S** **SBR** **BLOOD GAS** **HAEMATOCRIT** **IONISED CALCIUM** **TRUE GLUCOSE** **COAGULATION PROFILE**
COMPLETION OF EXCHANGE	**FBC** **U&E'S** **SBR** **BLOOD GAS** **IONISED CALCIUM** **HAEMATOCRIT** **TRUE GLUCOSE** **COAGULATION PROFILE**
EVERY HOUR THROUGH EXCHANGE	**WARD GLUCOSE TEST**
1, 2 AND 4 HOURS POST EXCHANGE	**WARD GLUCOSE TEST**

9. POST PROCEDURE CARE

9.1 Flush all lines with saline

9.2 Continue intensive phototherapy and repeat bilirubin levels in 2 hours

9.3 Continue with full monitoring for 6 hours post procedure

9.4 Inform parents that procedure has been completed

9.5 Observe for abdominal distension, vomiting or blood in stools

9.6 Perform urinalysis

9.7 Recommence enteral feeds 2 hours post transfusion

9.8 Dispose of all blood products as per trust policy

9.9 The procedure should be contemporaneously documented, in line with trust and national policy

10. COMPLICATIONS

- Cardiac arrhythmias
- Apnoea
- Acid base instability
- Circulatory overload
- Electrolyte disturbances i.e.: hypokalaemia, hypoglycaemia
- Embolism
- Hypothermia
- Infection
- Necrotising enterocolitis
- Thrombocytopaenia
- Intraventricular haemorrhage

11. MONITORING COMPLIANCE WITH THIS GUIDELINE

Audit of practice to ensure treatment of hyperbilirubinemia is appropriate.

Any concern or non-compliance with this guideline that is identified through the investigation of clinical incidents, complaints or claims will be reviewed as per Trust policy and may result in an audit and / or amendment to the guideline.

12. REFERENCES

NICE guidelines: Jaundice in newborn babies under 28 days 2010. Updated 2016

Blackburn S. (1995). Hyperbilirubinemia and Neonatal Jaundice: Neonatal Network October 1995. Vol 14 No 7 PP 15-25

Boyd S. (2004). Treatment of physiological and pathological jaundice. Nursing times March 2004 Vol 100 No.13

Joint United Kingdom Blood Transfusion and Tissue Transplantation Services Professional Advisory Committee (JPAC), 2021

13. ASSOCIATED DOCUMENTATION

Umbilical Artery Catheterisation (UAC) in Neonates
UMBILICAL VENOUS CATHETERISATION
Red Blood Cell Transfusion in Neonates

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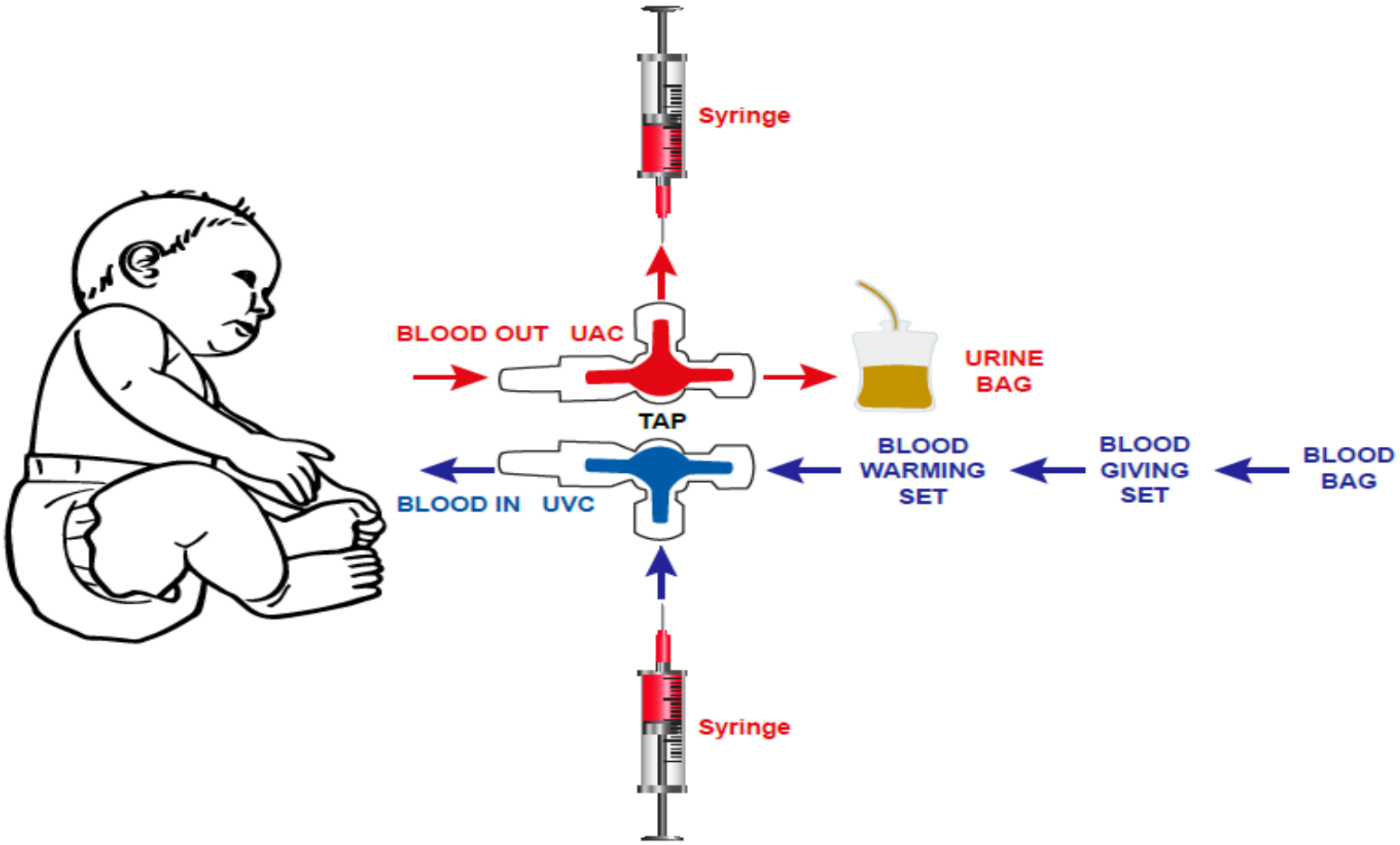
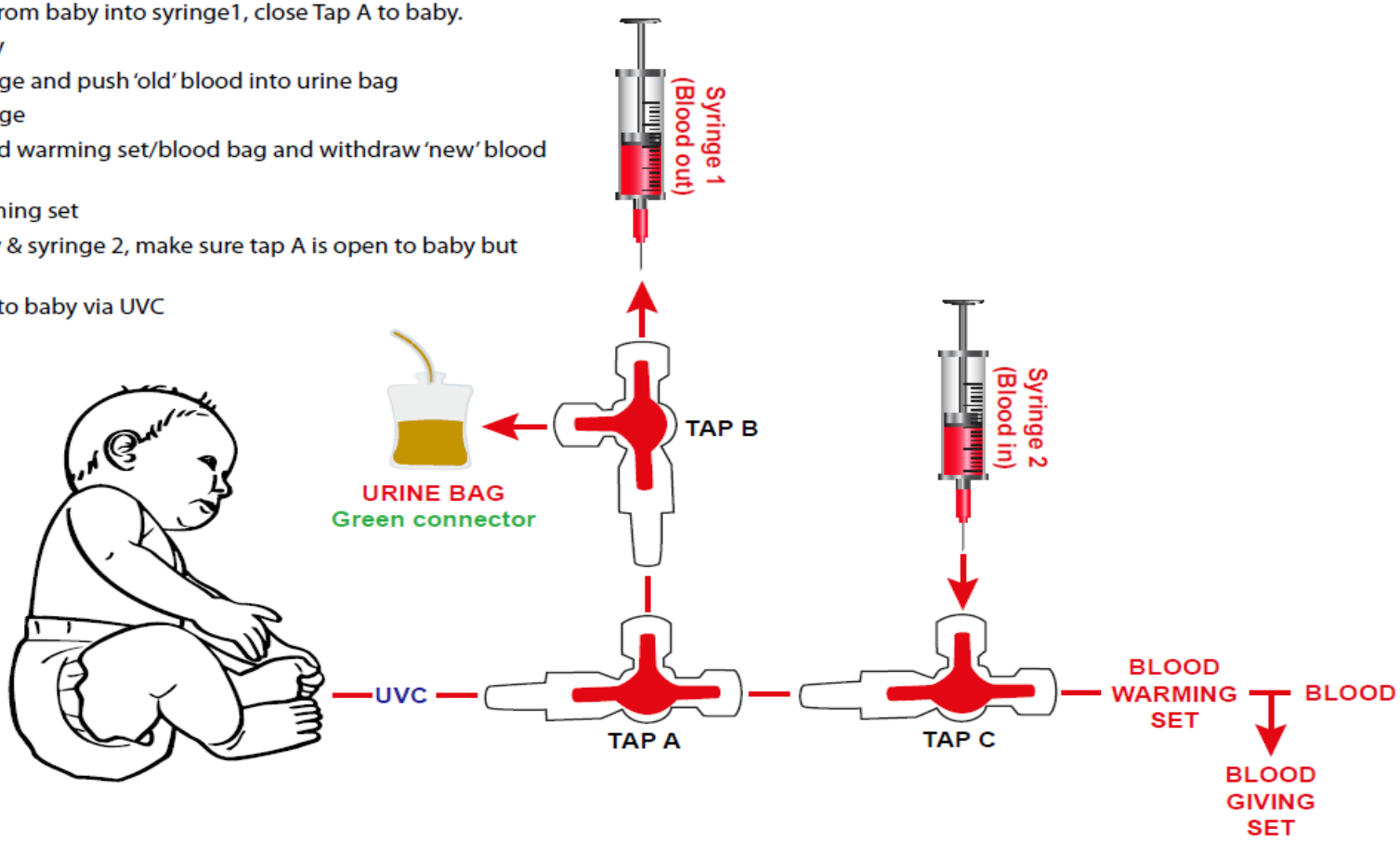
APPENDIX 1 – Please use SmartPhrase .nnuexchangetransfusion to record procedure on MyCare OR print EXCHANGE TRANSFUSION RECORDING CHART (to be completed and uploaded on Media).

APPENDIX 2
UAC – UVC EXCHANGE TRANSFUSION DIAGRAM



APPENDIX 3
SINGLE LINE EXCHANGE

Taking blood from baby

- Close Tap C to baby
- Close tap B to urine bag
- Open Tap A and B to baby
  – withdraw blood from baby into syringe1, close Tap A to baby.
- Close Tap B to baby
- Open Tap B to syringe and push 'old' blood into urine bag
- Close Tap A to syringe
- Open Tap C to blood warming set/blood bag and withdraw 'new' blood into syringe 2
- Close Tap C to warming set
- Open Tap C to baby & syringe 2, make sure tap A is open to baby but closed to syringe 1
- Push 'new' blood into baby via UVC